

Malhar Gudekar

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EDUCATION

University of Illinois Urbana Champaign

August 2024 - May 2026

Master of Science, Information Management

GPA: 3.8/4

• Coursework: Web Programming, Machine Learning & Cloud, Information Management, Information Consulting, Methods of Data Science

University of Mumbai

June 2019 - May 2023

Bachelor of Engineering, Computer Science

GPA: 9.25/10

• Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Object-Oriented Programming, Machine Learning, NLP

SKILLS

• **Languages & DSA:** Python, SQL, Data Structures & Algorithms (Arrays, Hashing, Trees, Graphs, Dynamic Programming), LeetCode

• **Backend & Systems:** FastAPI, REST APIs, ETL Pipelines, Kafka, PySpark, LangChain, Prompt Engineering, OpenAI API, MLOps

• **Databases, Cloud & Tools:** PostgreSQL, Neo4j, AWS (S3, EC2, EKS), GCP, Docker, Git

WORK EXPERIENCE

Business Intelligence Group

August 2025 - December 2025

Technical Consultant

Champaign, USA

- Built and deployed scalable multi-step agentic AI system with RAG, tool-use patterns, and LangChain orchestration using FastAPI, PostgreSQL, and vector embeddings for production applications
- Designed efficient, fault-tolerant **ETL and async pipelines** to process and index large document datasets for low-latency retrieval systems
- Improved system performance by **64%** through optimized indexing strategies and pipeline improvements in production environments
- Developed clean, production-ready backend services enabling real-time AI model serving, participating in code reviews to maintain quality and consistency across distributed systems

University of Illinois Urbana Champaign

May 2025 - Present

Research Assistant - iSchool

Champaign, USA

- Owned **end-to-end design, development, and integration** of cross-platform **AI backend system** ingesting wearable data for 100+ users, enabling scalable data collection and downstream analytics
- Designed and optimized **PostgreSQL schema, indexing strategies, and distributed query execution plans** for high-volume datasets, improving query performance by 38% in production workloads
- Built production-ready Python LLM pipeline with prompt engineering, retrieval, and context summarization, improving accuracy by 43%
- Developed asynchronous AI backend workflows with monitoring and CI/CD integration for real-time, fault-tolerant data processing with focus on production reliability

Research Assistant - CHI AI in Infodemic Management

January 2025 - May 2025

- Designed and deployed highly **scalable NLP and generative AI pipelines** in **Python** to process datasets, reducing processing latency by **41%**
- Built distributed data ingestion pipelines using **Kafka** and **Neo4j** Streams for real-time, high-throughput processing
- Designed and developed **production-grade REST APIs using FastAPI** to serve AI models, implementing validation, monitoring, and automated deployment workflows to ensure **scalable integration** across distributed systems
- Researched and evaluated emerging AI frameworks and tools, integrating relevant findings into team engineering workflows and technical solution design

Swift Mobil Software Solutions Provider

July 2023 - October 2023

Data Analyst

Mumbai, IN

- Built interactive **Power BI** dashboards for **logistics** and **mobility KPIs**, reducing reporting time by **40%** and improving operational visibility
- Processed and optimized large-scale datasets using **Python** and **PySpark** to identify failure and performance issues in distributed systems
- Developed **SQL** and **graph queries** to map dependencies, analyze operations, detect data issues, speeding root-cause analysis by **26%**

Pscope Technologies Pvt. Ltd.

January 2023 - June 2023

Data Analyst

Mumbai, IN

- Developed and maintained **20+ Power BI** dashboards across Sales, Finance, and Operations for enterprise clients, leveraging **SQL** and **DAX**, enabling KPI tracking and data-driven decisions
- Automated recurring **Excel**-based workflows using **VBA**, implementing validation and scheduling logic, reducing report preparation time by **42%** and improving consistency

Stu/Dio at Illinois

August 2025 - Present

Project Manager

Champaign, USA

- Collaborated with cross-functional teams of engineers and designers to deliver AI-driven features for a consumer-facing product, using engagement metrics and user feedback to guide decisions
- Tracked **sprint** progress, delivery metrics, and **QA issues** through **JIRA**-based workflows, managing backlog and issue resolution

PROJECTS

AI-Powered AR Virtual Try-On (Glossier)

[View Project](#)

- Conducted customer segmentation and behavioral analysis on beauty retail users, identifying high engagement trends among users aged 18–44 for AR-based product experiences
- Designed AI-assisted personalization concepts for virtual try-on workflows, mapping recommendation logic and user interaction flows to improve engagement and conversion potential
- Developed strategic roadmap outlining AR integration, ROI projections, and KPIs including return reduction, session retention, and conversion improvement across digital commerce channels